

choptuik_ac_bc · theorems12 ·
github.com/wild8highlander/choptuik_ac_bc

The Honest-Negative Theorem

Cluster: Not DSS, the Mass Floor, and the Linear Certificate

$\Delta_{\text{eff}} = 0.611 \pm 0.446$ (3.44 excluded at 6.3σ) · $\gamma = 0.109 \pm 0.105$ · $\Pi = \text{I} + \text{B} \cdot \text{T}$
to $z = 30$

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Machine:	choptuik_scaling + amplitude_ladder + depth_z30 + deep_echo_v22
Verdict file:	results/choptuik_scaling.json + v20b_depth_z30.json + v22_v_budget.json

Abstract

The final monograph of the cycle records the honest-negative theorem cluster of the verification programme: (a) the quartet period of the deep run $\Delta_{\text{eff}} = 0.6114 \pm 0.4459$ is honestly not DSS — the literature period 3.44 is excluded at 6.3σ and the estimate is compatible with zero; (b) the fixed-grid mass-scaling campaign yields $A^* = 0.077744$ and $\gamma = 0.109 \pm 0.105$ against the anchor 0.374, an honest negative traced to the mass floor $M \approx 4 \cdot \text{du}$ at horizon nucleation; (c) echo trains do not develop on any saved campaign, so the finite-amplitude exponent p is not measurable on the current machine — the discriminator $p = 1/2$ vs 1 vs 2 is preserved as a falsifiable protocol with a $\sim 15\text{--}20$ s per-rung cost.

(d) The linear sector is certified to $z = 30$ in closed form: $\Pi(T) = I + B \cdot T$ with phase exactly zero and secular growth ~ 126 ; the floor budget $S_{\text{req}}(30) \approx 2.26 \times 10^3$ ($\times 10$ margin) quantifies the measurement requirement, and the percentile ladder delivers percent-level γ/Δ (5/5) where naive OLS fails (-23%). Negative results are first-class: they bound the interpretation of every positive theorem of the series. Verdict tables, corollaries, honest caveats and the reproduction protocol are included.

Keywords: honest negatives, Δ_{eff} not DSS, mass floor, amplitude ladder, p -discriminator, linear certificate, machine theorems.

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1. Introduction

The final monograph of the cycle records the honest-negative theorem cluster: the machine results of the programme that bound what the current machines can and cannot see. Four results are fixed: (a) the quartet period $\Delta_{\text{eff}} = 0.611 \pm 0.446$ of the deep run is honestly NOT DSS — the literature period 3.44 is excluded at 6.3σ and the estimate is compatible with zero; (b) the fixed-grid mass-scaling exponent $\gamma = 0.109 \pm 0.105$ against the literature anchor $\gamma_{\text{Ch}} = 0.374$ is an honest negative traced to the mass floor $M \approx 4 \cdot \text{du}$ at horizon nucleation; (c) the echo train does not develop on any saved campaign (0–3 peaks, $z \leq 6.05$), so the finite-amplitude exponent p is not measurable on the current machine — the discriminator $p = 1/2$ versus 1 versus 2 is preserved as a falsifiable protocol; (d) the linear sector is certified to $z = 30$ in closed form (phase exactly 0, secular growth ~ 126), so any genuine DSS signal must be finite-amplitude.

Negative results are first-class in this repository: they bound the interpretation of every positive result of the series and protect the programme from reading artifacts as physics. The cluster is stated as theorems in the machine sense: each claim is tied to a deterministic verdict file and to tests.

2. Notation and Setting

The critical collapse programme compares against the anchors $\gamma_{\text{Ch}} = 0.374$ (Choptuik) and $\Delta \approx 3.44$ (the CSS echo period), and against the DSS normalization $\Delta = 6\ln(16/9) = 3.452$. The mass-scaling law reads $M = C(A - A^*)^\gamma$ near the critical amplitude A^* ; the fixed-grid campaign bisects A^* and fits γ over the echo-resolving ladder. The deep-echo quartet Δ_{eff} is the spacing of the four Q-peaks of one stage; the discriminator p concerns the scaling of the echo frequency with amplitude: $p = 1/2$ predicts a Jordan-2 carrier, $p = 1$ an exit-type carrier, $p = 2$ a Lyapunov-center carrier.

The certificate machine depth_{z30} closes the linear sector: the propagator over an echo is $\Pi(T) = I + B \cdot T$ with B nilpotent (monograph No. 5), hence the phase is exactly zero and the growth is secular; the floor-suppression budget $S_{\text{req}}(z) = e^{(\gamma \cdot 20.65)}$ quantifies what a genuine measurement at depth z requires.

3. Statement of the Theorem

Theorem 12 (the honest-negative cluster). On the current machine class of the repository:

(a) the quartet period of the deep run is $\Delta_{\text{eff}} = 0.6114 \pm 0.4459$ (stage 4); the DSS value 3.44 is excluded at 6.3σ and the estimate is compatible with zero: DSS periodicity is honestly not confirmed at $z \leq 8.56$;

(b) the fixed-grid campaign yields $A^* = 0.077744$ (bisection, $n = 3000$) and $\gamma = 0.10903 \pm 0.10534$ ($C = 0.009327$, $r^2 = 0.565$) against the anchor $\gamma_{\text{Ch}} = 0.374$; the deviation is traced to the mass floor $M \approx 4 \cdot \text{du}$ at horizon nucleation — a documented machine limitation, not hidden;

(c) echo trains do not develop on any saved campaign (0–3 Q-peaks per chain, $z \leq 6.05$ across all saved runs; ≤ 3 peaks on the v21/v22 second passes): the exponent p is not measurable on this machine; the discriminator (1/2, 1, 2) and the per-rung cost (~ 15 – 20 s) are recorded as a falsifiable protocol;

(d) the linear sector is certified to $z = 30$: $\Pi(T) = I + B \cdot T$, phase exactly 0, secular growth ~ 126 over 41 spinor echoes; the floor budget $S_{\text{req}}(30) = e^{(\gamma \cdot 20.65)} \approx 2.26 \times 10^3$ ($\times 10$ margin) quantifies the measurement requirement; the closed loop delivers percent-level γ/Δ on the reduced ladder (5/5 scenarios) while naive OLS at the wall reproduces the q3 failure (-23%).

4. Sketch of the Machine Proofs

4.1. The quartet and the DSS exclusion (a)

The four Q-peaks of the deep run's stage 4 are fitted for a common spacing; the estimate and its uncertainty come from the stage's peak ladder. The DSS hypothesis $\Delta = 3.44$ lies 6.3 estimated standard deviations away and is excluded; the estimate itself is compatible with zero spacing, which is why the honest reading is “sub-echo structure or floor-contaminated ladder” rather than “small DSS”. The second-pass amplitude ladders (v21, v22) find no Δ_{eff} on at least three rungs, reinforcing the negative.

4.2. The mass floor (b) and the p-discriminator (c)

The fixed-grid machine bisects A^* to 0.077744 and fits the scaling on the ladder points; the fit's large uncertainty and the plateau $M \approx 4 \cdot du$ at horizon nucleation explain the deviation from the anchor: the machine cannot resolve masses below the floor, and the fitted γ mixes the true scaling with the floor. The amplitude ladders (v20a, and the second passes) execute the falsifiable protocol: each rung costs $\sim 15\text{--}20$ s, the stop conditions are recorded (singularity or $v_{\text{exhausted}}$), and no rung develops an echo train — so p is not measurable, and the discriminator stays falsifiable for a deeper machine rather than being silently dropped.

4.3. The linear certificate (d)

The certificate machine propagates the linear sector in closed form: with B nilpotent, $\Pi(T) = I + B \cdot T$ is exact, the phase is exactly zero (no linear monodromy), and the secular growth reaches ~ 126 by $z = 30$. The budget $S_{\text{req}}(30)$ follows from the wall arithmetic; the percentile ladder closes the loop — with the budget applied, the a-priori estimator rule delivers percent-level γ/Δ (5/5), while naive OLS at the wall reproduces the -23% failure of q_3 . This is the measurement discipline that converts the negatives of (a)–(c) into a quantified requirement for the next machine generation.

Table 12.1. The cluster: verdict files and headline numbers

Claim	Machine / verdict	Headline
(a) not DSS	deep_echo_v22 (C3)	$\Delta_{\text{eff}} = 0.6114 \pm 0.4459$; 3.44 excluded at 6.3σ
(b) mass floor	choptuik_scaling	$A^* = 0.077744$; $\gamma = 0.109 \pm 0.105$ vs 0.374; $M \approx 4 \cdot du$
(c) p not measurable	amplitude_ladder + second passes	0–3 peaks, $z \leq 6.05$ saved; rung cost $\sim 15\text{--}20$ s
(d) linear certificate	depth_z30	$\Pi = I + B \cdot T$; phase 0 exactly; $S_{\text{req}}(30) \approx 2.26 \times 10^3$

5. Numerical Results

Table 12.2. Fixed-grid scaling fit (choptuik_scaling.json)

Quantity	Value
A* (bisection, n = 3000)	0.07774376869201663
γ (fit)	0.10903361370709194
γ (standard error)	0.10534278715730508
C (amplitude)	0.009327287284718152
r^2	0.5654649659717386
Mass floor	$M \approx 4 \cdot du$ at horizon nucleation
Anchor γ_{Ch} (comparison only)	0.374

Table 12.3. The echo-train programme across campaigns (peak counts per chain)

Campaign	Depth z	Q-peaks	Stop
v20a rungs	≤ 4.6	0–3	singularity / v_exhausted
saved campaigns (all)	≤ 6.05	≤ 3	—
v21 w = 3	7.20	3	v_exhausted
v22 deep (k1 = 0.52)	6.99	7 (4 on stage 4)	v_exhausted

The certificate arithmetic of (d): $S_{req}(30) = e^{(0.374 \cdot 20.65)} \approx 2.26 \times 10^3$, with a tenfold margin 2.3×10^4 ; bisection cannot carry the depth ($\epsilon \sim 10^{-35}$), zoom rescaling does — which is precisely why the wall theorem (monograph No. 11) governs the frontier. The percentile ladder’s 5/5 versus OLS’s –23% at the wall is the quantitative form of the lesson: estimators must be matched to the budget regime.

6. Corollaries

Corollary 12.1 (no DSS claim at current depth). Until a machine reaches the budget regime, every echo-period statement of the programme must be phrased as compatible-with-zero; the DSS normalization $6\ln(16/9) = 3.452$ remains a comparison anchor only. Any published claim to the contrary would contradict the recorded verdicts.

Corollary 12.2 (the floor explains the scaling gap). The gap between $\gamma = 0.109 \pm 0.105$ and the anchor 0.374 is a resolution statement ($M \approx 4 \cdot du$), not a physics statement: the fixed-grid ladder cannot distinguish sub-floor masses. The zoom programme inherits the measurement burden, and the budget (d) is its quantified form.

Corollary 12.3 (the discriminator is falsifiable, not abandoned). The p-discriminator (1/2, 1, 2) with its per-rung cost is recorded for the next machine generation: it will either measure p at depth or fail visibly — both outcomes are informative, and the cluster forbids neither.

7. Honest Caveats

The quartet Δ_{eff} is measured on one stage of one chain at Q-levels of 10^{-12} – 10^{-11} — floor contamination is plausible and the sub-echo interpretation is open; the 6.3σ exclusion applies to the DSS value, not to any specific alternative. The mass-floor explanation of (b) is a documented mechanism, not a fitted correction: the fit's $r^2 = 0.565$ shows the ladder is only partially in the scaling regime. The p-not-measurable verdict is machine-specific: a deeper machine may develop trains, and the discriminator is preserved for exactly that case. The certificate (d) is linear-sector exact: nonlinear echoes, if any, are outside its scope, and the budget assumes the wall arithmetic of monograph No. 11. All anchors — γ_{Ch} , Δ , Δ_{sp} — are comparison anchors; nothing is fitted to them.

8. Reproducibility

Table 12.4. Reproduction protocol for Theorem 12

Step	Command	Output
1	python3 choptuik_scaling.py --n-bisect 1200	results/choptuik_scaling.json (A^* , γ fit, floor)
2	python3 amplitude_ladder.py	results/v20a_amplitude_ladder.json (~1–2 min)
3	python3 depth_z30.py	results/v20b_depth_z30.json (~1 min; certificate + budget)
4	python3 deep_echo_v22.py --phase all	results/v22_v_budget.json (quartet Δ_{eff})
5	python3 -m pytest tests/ -q	58 passed (v20–v22 tests included)

All machines are deterministic and single-threaded; runtimes are minutes on a laptop and 2–3× that in Termux. The data correspond to commit 5b256cd; self-contained copies live in `einstein_direct/open9/machines/` with a private results copy.

References

- [1] Repository `wild8highlander/choptuik_ac_bc`: `einstein_direct/choptuik_scaling.py`;
`einstein_direct/results/choptuik_scaling.json`.
- [2] Ibid.: `einstein_direct/amplitude_ladder.py`; `einstein_direct/results/v20a_amplitude_ladder.json`
(campaign v20a).
- [3] Ibid.: `einstein_direct/depth_z30.py`; `einstein_direct/results/v20b_depth_z30.json` (campaign v20b).
- [4] Ibid.: `einstein_direct/deep_echo_v22.py`; `einstein_direct/results/v22_v_budget.json` (campaign v22,
quartet).
- [5] Ibid.: `einstein_direct/README.md` §§1.2, 3.9–3.11.
- [6] Monographs No. 5, No. 11 of the theorems12 cycle.
- [7] M. W. Choptuik. Phys. Rev. Lett. 70, 9 (1993) — the $\gamma_{\text{Ch}} = 0.374$ comparison anchor.